

! Addressing on Networks

There are four main addressing methods used in networks, each connected to a different OSI layer:

- **Application layer:** Uses **FQDNs (fully qualified domain names)**, computer names, or host names to identify devices.
- **Transport layer:** Uses **port numbers** to identify specific services or applications.
- **Network layer:** Uses **IP addresses** (IPv4 has 32 bits and uses 4 decimal numbers; IPv6 has 128 bits and uses 8 hexadecimal blocks).
- **Data Link layer:** Uses **MAC addresses** (also known as physical addresses) to identify devices on a local network.

These addresses help data reach the correct device and application in the network.

MAC Addresses

A traditional MAC address is made of **two parts**:

- The **first 24 bits** are called the **OUI (Organizationally Unique Identifier)**, which is assigned by **IEEE** to the company.
- The **last 24 bits** are the **device ID**, a unique number given by the **manufacturer** to each network card ((NIC)(Network Interface Card))

Together, they create a **unique hardware address** for every network device.



Figure 3-1 NIC with MAC address

(DNS and functions of it)

! The Domain Name System (DNS) !

The **Domain Name System (DNS)** is an Application layer client-server system **used to translate computer names into IP addresses**. It includes three main parts:

- **Namespace:** A global collection of names and their related IP addresses stored in DNS databases.
- **Name servers:** Special servers that store these databases in a hierarchical structure.
- **Resolvers:** DNS clients that ask name servers for information.

DNS helps devices find each other on the internet by converting human-readable names (like example.com) into machine-readable IP addresses.

DNS name servers are arranged in a hierarchical structure.

- At the **top level**, there are 13 root server clusters that help find **top-level domain (TLD)** servers (like .com, .org).
- **TLD servers** then provide the location of **authoritative servers**, which store the actual names and IP addresses for computers within their domain.

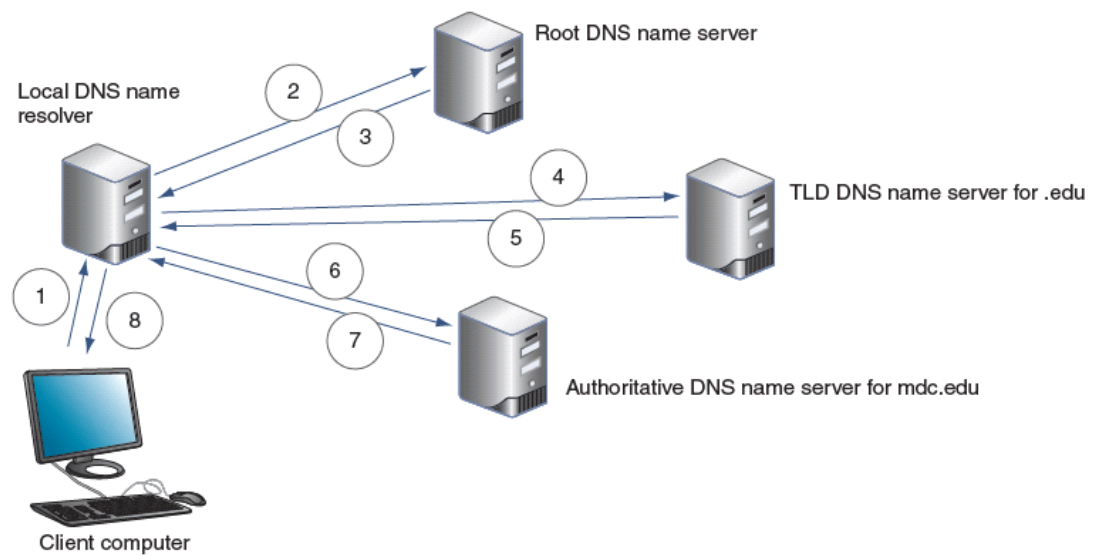


Figure 3-18 Queries for name resolution of `www.mdc.edu`

How DNS Servers Work

User types "www.mdc.edu" in the browser

[1] Client checks the local **DNS cache**

- If the IP address is already stored, the website opens immediately.

→ If IP is found → Direct access

- If not found, a DNS query starts to find the IP.

[2-3] Local DNS server sends a request to the Root DNS server

- The Root DNS server doesn't know the final IP, but it tells the local DNS where to find the TLD (.edu) server.

[4-5] Local DNS contacts the .edu TLD DNS server

- The TLD server doesn't know the full IP address, but it gives the address of the authoritative DNS server for mdc.edu.

[6-7] Local DNS sends a request to the authoritative DNS server for mdc.edu

- The authoritative server knows the exact IP address for www.mdc.edu and sends it back to the local DNS server.

[8] Local DNS server sends the IP address back to the client computer

- The client stores (caches) the IP to avoid asking again soon.

👉 IP is cached → Website is loaded in the browser

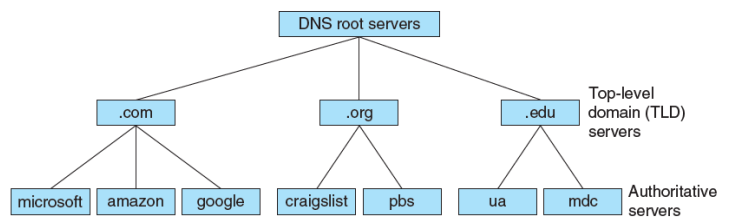


Figure 3-17 Hierarchy of name servers

Sometimes DNS resolution is more complex:

- A **caching-only server** does not store full DNS records. **Caching-only servers** ask the company's authoritative server if the info is not in cache.
- Some internal name servers can't access **root servers** directly.
- A **TLD server** may point to an **intermediate server** instead of the final one.

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There are two main types of **DNS queries**:

- **Recursive query:**

The client asks the local DNS server to fully resolve the domain name or return an error if it can't be found. The local server does all the work and gives the final answer.

- **Iterative query:**

The local DNS server asks other servers step-by-step for information. Each server only answers if it has the data. If not, it points to another server. The local server collects and returns the answer.

Socket = IP address + Port Number

(socket meaning and examples of that)

Ports and Sockets

- **Port numbers** make sure data goes to the correct application on a device.
- A **Socket** combines an IP address and a port number (e.g., 10.43.3.87:23) to identify a specific connection between two systems.
- **Port numbers** are grouped into three types:
 - **Well-known ports** (0–1023) – used by common services like HTTP or FTP
 - **Registered ports** (1024–49151) – used by user or vendor applications
 - **Dynamic/private ports** (49152–65535) – usually for temporary connections

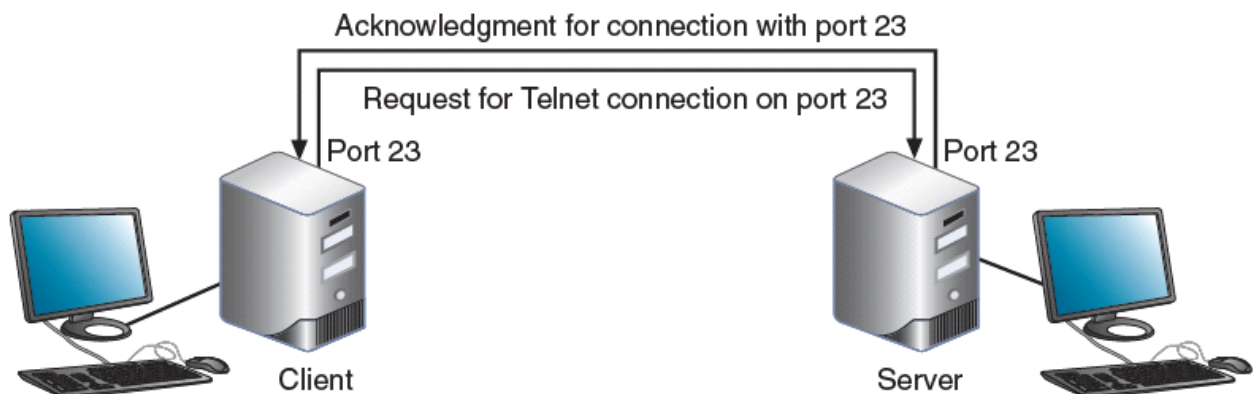


Figure 3-16 A virtual connection for the Telnet service

[IP addresses. Explained.](#)

There are two types of **IP addresses**: (classes of ip addresses)

- **IPv4** uses 32 bits and is written in four decimal numbers (e.g., 192.168.0.1).
- **IPv6** uses 128 bits and is written in eight groups of hexadecimal numbers (e.g., 2001:0db8:85a3::8a2e:0370:7334).

Classes of IP Addresses

Table 2-4 IP address classes

Class	Network octets*	Approximate number of possible networks or licenses	Approximate number of possible IP addresses in each network
A	1.x.y.z to 126.x.y.z	126	16 million
B	128.0.x.y to 191.255.x.y	16,000	65,000
C	192.0.0.x to 223.255.255.x	2 million	254

*An x, y, or z in the IP address stands for an octet that is used to identify hosts.

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How IPv4 Addresses Are Formatted and Assigned ?

IPv4 addresses are 32-bit numbers divided into four groups of 8 bits, called **octets**. Each octet can range from 0 to 255. Some addresses are reserved for special uses

IPv4 addresses are also divided into **five classes**:

- **Class A**: Uses the **first octet** for the network (**for large networks**)
 - **Class B**: Uses the **first two octets** (for **medium-sized** networks).
 - **Class C**: Uses the **first three octets** (for **small** networks).
 - **Class D**: Used for **multicasting** (224-239).
 - **Class E**: Used for **research** (240-254).
- (Class D,E were not available for general use.)

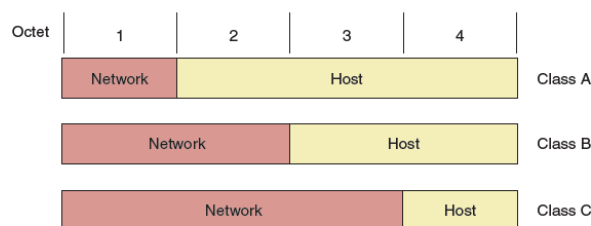


Figure 3-5 The network portion and host portion for each class of IP addresses

- Public IP addresses (Class A, B, and C) are **used** on the Internet.
- Companies can use **private IP addresses** for internal networks. IEEE recommends the following IP addresses be used for private networks:
10.0.0.0 through 10.255.255.255
172.16.0.0 through 172.31.255.255
192.168.0.0 through 192.168.255.255

How a DHCP Server Assigns IP Addresses

There are two types of IP address assignments:

- **Static IP addresses** are assigned manually by a network administrator.
- **Dynamic IP addresses** are automatically assigned by a **DHCP server**.
If the DHCP server is not available, the computer uses an **APIPA** address from the range **169.254.0.1 to 169.254.255.254**.

Address Translation, NAT, and PAT

- NAT (Network Address Translation) NAT is a system that translates private IP addresses into public IP addresses when accessing the internet.
- PAT (Port Address Translation) is a type of NAT that adds a unique port number to each session, so multiple devices can use the same public IP at once.

NAT is a method used by networks to save public IP addresses. When a computer in a private network wants to connect to the Internet, a special device (called a gateway) replaces its private IP with the network's one public IP. This process is called *address translation*.

NAT is useful because:

- The network only needs **one** public IP.
- It **protects** private devices, hiding them behind the public IP.

So, NAT helps both with **IP address saving** and **security**.

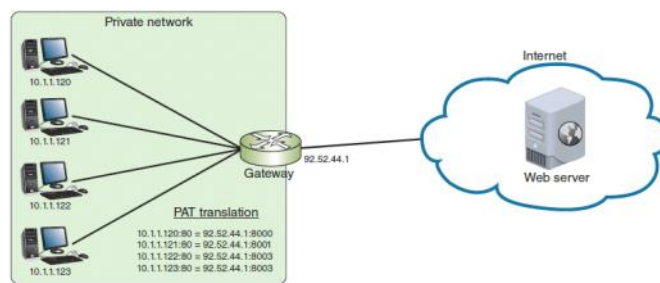


Figure 3-9 PAT (Port Address Translation)

How IPv6 Addresses Are Formatted and Assigned

IPv6 addresses are 128-bit long and written as eight groups of hexadecimal numbers separated by colons. Each block is 16 bits. Leading zeros in a block can be removed, and if multiple blocks contain only zeros, they can be replaced with "::", but only once in an address.

Example: 2001:0000:B80::D3:9C5A:CC

IPv6 Terminology

- **Link:** A local network area bounded by routers.
- **Interface:** A device's connection to a network.
- **Tunneling:** A method to send IPv6 packets over an IPv4 network.
- **Interface ID:** The last 64 bits of an IPv6 address, used to identify a device.
- **Neighbors:** Devices on the same local network.

Types of IP Addresses



- **Unicast address:** Targets a single device.
 - **Global unicast:** Can be routed on the Internet.
 - **Link-local unicast:** Used within the same local network.
- **Multicast address:** Sends packets to all devices on the network.
- **Anycast address:** Sends packets to the nearest device among several options.

CHAPTER 3 Addressing on Networks

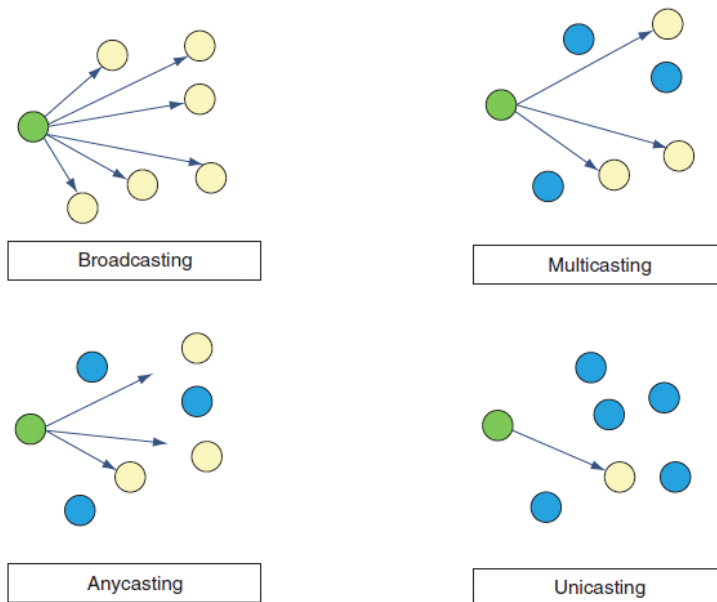


Figure 3-14 Concepts of broadcasting, multicasting, anycasting, and unicasting

Broadcasting

A message is sent from one device to all devices on the network.

Multicasting

A message is sent from one device to a specific group of devices.

Anycasting

A message is sent from one device to the nearest device in a group.

Unicasting

A message is sent from one device to only one specific target device.

Tunneling

When a network uses both IPv4 and IPv6 protocols, it is called **dual stacked**.

If data needs to pass through networks that don't support both protocols, **tunneling** is used to send IPv6 data over IPv4 networks.

There are three common tunneling protocols:

- **6to4**
- **ISATAP** (Intra-Site Automatic Tunnel Addressing)
- **Teredo**

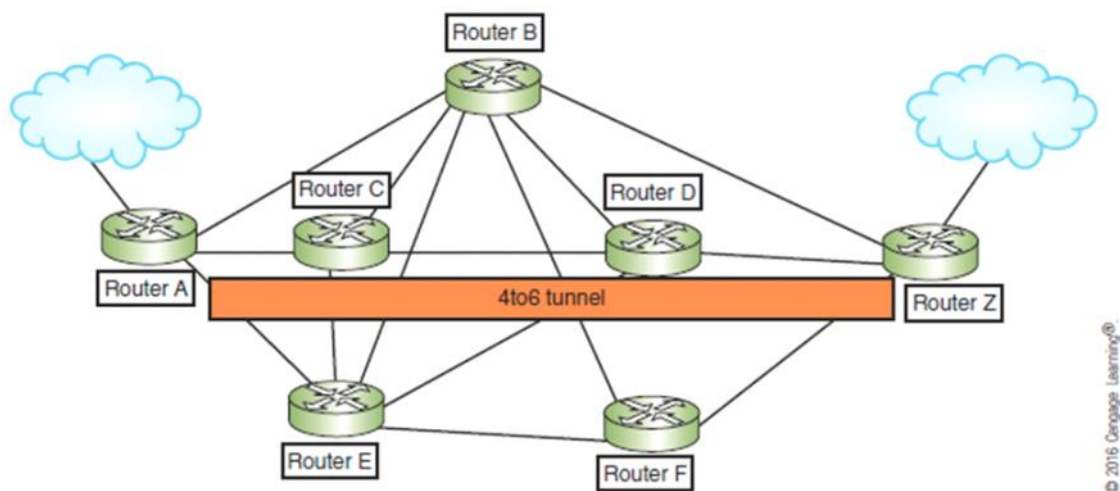


Figure 2-21 A 4to6 tunnel is used to move IPv4 packets through a futuristic IPv6 network that is configured to not use IPv4

SUMMARY

- Every device (host) on a network is given a **host name**, which helps identify it in a human-readable way.
- **Applications** use specific **port numbers** to communicate with other applications over the network. Each type of communication (like web, email, etc.) has its own port number.
- An **IPv4 address** is a 32-bit number divided into **four decimal numbers** (called **octets**), for example: 192.168.1.1.
- An **IPv6 address** is a 128-bit number written as **eight groups of hexadecimal values**, for example: 2001:0db8:0000:0000:0000:ff00:0042:8329.
- Every **NIC** (Network Interface Card) has a **unique 48-bit MAC address**. This is the hardware address used on local networks.
- You can use the **ipconfig command** on Windows to check your computer's IP address and other network settings.